

**GOAL:** Aim 2 of my project is to grow different cell types (B cells, astrocytes, and microglia) that are found in the brain together, expose them to things that cause Alzheimer's, and see how they "talk" to each other. Talking in biology looks like the sending of proteins called cytokines and chemokines. I'm not actually measuring the amounts of those proteins (yet) but I'm quantifying the *genes* that express them. For my Aim 1, I got the genes being expressed by B cells. Those genes are in the file I sent you (we care about the "beta" term for quantifying expression). To show I'm making progress on my Aim 2, which involves cellular communication, I want to show that I've looked into what chemokines and cytokines are being secreted (expressed) by B cells before I see what all the cells are doing together.

This guide hands off modeling intercellular communication between B cells, astrocytes, and microglia after neurodegenerative exposure. You will be given a list of cytokine/chemokine-related genes and their differential expression (DEG) results from a screen already completed in one of the three cell types (B cells, exposed alone). **Your job is to turn that gene list into a set of predicted signaling interactions with the other two cell types, and to produce the visualizations described in Section 5.**

## 2. Background: key terms

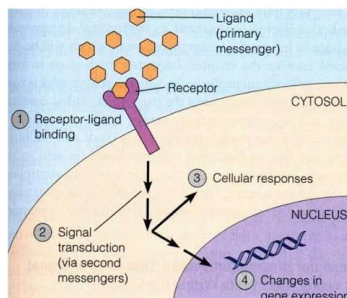
**Cytokine** — a small secreted signaling protein used by cells (especially immune cells) to communicate with each other. Cytokines regulate inflammation, immune activation, proliferation, and cell survival.

**Chemokine** — a subtype of cytokines whose main job is to induce directed cell migration (chemotaxis). Chemokines are classified by the spacing of conserved cysteine residues into four families: CC, CXC, CX3C, and C.

**Ligand** — in this context, any secreted or membrane-bound molecule (a cytokine, chemokine, or other signaling factor) that binds a receptor on a target cell to trigger a response. A gene only counts as a ligand candidate if its protein product is actually secreted or membrane-displayed — being differentially expressed is not sufficient on its own.

**Receptor** — a protein, usually transmembrane, expressed on a target ("receiving") cell that binds a specific ligand and transduces the signal into a change in that cell's behavior (gene expression, migration, activation state, etc.).

**Ligand–receptor pair** — the basic unit of predicted intercellular communication. If cell type A expresses/secretates ligand X, and cell type B expresses receptor Y that binds X, that pair is a candidate signaling event between A and B.



*Note: some entries in the gene list will be modulators or co-receptors rather than clean ligands or receptors — flag anything ambiguous rather than forcing it into one category.*

### 3. Biological context about chemokines

Chemokine signaling axes (e.g., CXCR4–SDF1/CXCL12, CX3CR1–CX3CL1, CXCR6–CXCL16) are established mediators of crosstalk between microglia, astrocytes, and neurons in both injury and neuroprotective contexts, and **dysregulated cytokine/chemokine signaling between glial cells is considered a central driver of chronic neuroinflammation in neurodegenerative disease**. This is the rationale for the project: **if the neurodegenerative exposure altered cytokine/chemokine expression in B cells, the next question is whether those changes are positioned to signal to astrocytes and/or microglia through matching receptors — and Aim 2 is designed to test that directly.**

### 4. Where the project stands

The overall Aim 2 workflow has six stages. Stages 1–2 are complete; you are picking up at stage 3.

- **Exposure screen** — completed. B cells exposed to a panel of neurodegenerative stressors, bulk DEG analysis performed.
- **DEG list** — completed. Cytokine/chemokine-related DEGs filtered and categorized (ligand, receptor, modulator, etc.) — this is the gene list and results you will receive.
- **Predicted ligand–receptor network** — your task. Match candidate secreted ligands from the B cell DEG list to receptors expressed by astrocytes and microglia.
- **Co-culture** — planned, not yet started. All three cell types cultured together.
- **scRNA-seq readout** — planned. Single-cell sequencing of the co-culture to measure actual transcriptional responses in astrocytes and microglia.
- **Validation** — planned. Compare observed co-culture interactions back against the network you predict in stage 3.

### 5. Step-by-step workflow (Stage 3)

Step 1 — Sort the gene list into ligands, receptors, and modulators

- Start from the category labels already assigned in the heatmap/DEG results, but verify them.
- Confirm each candidate ligand is annotated as secreted (or membrane-bound signaling molecule) using UniProt subcellular location or the Human Protein Atlas secretome annotation — do not assume a gene is a ligand just because it is in the cytokine/chemokine gene list.
- Confirm each candidate receptor is annotated as transmembrane.
- Set aside anything ambiguous (modulators, soluble receptor decoys, co-factors) for discussion rather than forcing a category.

Step 2 — Assemble receptor-expression reference data for astrocytes and microglia

- Since we have not yet generated astrocyte/microglia expression data ourselves, pull baseline expression from a public reference (e.g., a relevant single-cell atlas of human or mouse glia, or the Human Protein Atlas).
- Record, for every receptor gene from Step 1, whether it is expressed in astrocytes, microglia, both, or neither.

### Step 3 — Run ligand–receptor interaction prediction

Use a curated ligand–receptor interaction tool rather than pairing genes manually. Two standard options:

- **CellPhoneDB** — matches expression of curated multi-subunit ligand–receptor complexes across cell populations; a good default for a straightforward "which pairs are plausible" analysis given our current data (B cell DEGs + public receptor expression for astrocytes/microglia).
- **CellChat** — incorporates multi-subunit receptor structure plus soluble/membrane-bound cofactors, and has built-in functions for circle and chord-diagram visualization directly from its output; a good option if a polished summary figure is needed quickly.
- **NicheNet** — infers which ligands best explain observed expression changes in the receiving cell. This becomes more useful in Stage 5, once real astrocyte/microglia scRNA-seq data exists post-co-culture, since it needs receiver-side transcriptomic response data to rank ligands, not just receiver-side expression.
  - People in my lab have used this before

*For this stage, run CellPhoneDB and/or CellChat on the B cell ligand list against the astrocyte and microglia receptor-expression reference from Step 2. Save NicheNet for the post-co-culture scRNA-seq analysis in Stage 5.*

### Step 4 — Generate the visualizations

See the checklist in Section 6 for the specific plots expected from this stage.

### Step 5 — Flag priority candidates

- From the predicted network, identify the strongest/most confident ligand–receptor pairs (highest DEG significance in B cells + confirmed receptor expression in astrocytes and/or microglia).
- These become the leading hypotheses to test once the co-culture and scRNA-seq stages begin — bring a short prioritized list (5–10 pairs) to the next lab meeting.

## 6. Deliverables checklist

| Plot                                    | What it shows                                                                                                                                                                                                                      | Source                                                    |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| Category-level summary bar chart        | Number of significant DEGs per exposure condition, grouped/stacked by category (ligand, receptor, modulator, chemokine, etc.). Quantifies how strongly each exposure perturbed the B cell secretome.                               | B cell DEG results (provided)                             |
| Ligand–receptor dot plot                | Ligand–receptor pair (rows) x cell-type pair, e.g. B cell→astrocyte, B cell→microglia (columns); dot size = interaction strength/significance, color = expression level. Standard CellPhoneDB-style output.                        | CellPhoneDB / CellChat output                             |
| Bipartite ligand–receptor network graph | Ligands (B cells) on one side, receptors (astrocytes/microglia) on the other, edges = predicted interaction, node size/color = fold-change or significance. Shows specific genes driving the network, not just aggregate strength. | Built in Cytoscape / igraph / networkx from Step 3 output |
| (Optional) circle/chord diagram         | Cell types as segments around a circle, ribbons sized by number/strength of predicted interactions. Fast, high-level summary — useful for meetings, less useful for identifying specific genes.                                    | CellChat netVisual_circle / netVisual_chord_cell          |

## 7. Key references

- Efremova, M., Vento-Tormo, M., Teichmann, S. A. & Vento-Tormo, R. CellPhoneDB: inferring cell–cell communication from combined expression of multi-subunit ligand–receptor complexes. Nat. Protoc. 15, 1484–1506 (2020).
- Jin, S. et al. Inference and analysis of cell–cell communication using CellChat. Nat. Commun. 12, 1088 (2021).
- Browaeys, R. et al. Unraveling cell–cell communication with NicheNet by inferring active ligands from transcriptomics data. Nat. Protoc. (2025).
- Frontiers in Immunology. Crosstalk Between Astrocytes and Microglia: An Overview (2020) — background on secreted-mediator crosstalk between glial cell types.
- PMC10740299 — Bruton's tyrosine kinase-bearing B cells and microglia in neuromyelitis optica spectrum disorder — direct evidence of B cell–microglia chemokine-mediated crosstalk.
- Cell/Neuron 2025 (BST2/C3–C3aR paper) — example of CellPhoneDB applied specifically to astrocyte–microglia ligand–receptor inference.

*Questions on any step should go back to [PI/your name] before finalizing figures — this is a living workflow and may be adjusted once real astrocyte/microglia expression data or early co-culture results come in.*